Anomaly Detection in Hyperspectral Imagery Based on Gaussian Mixture Model

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***Abstract*— Hyperspectral images (HSIs) with rich spectral information have been widely used in many fields. Anomaly detection is one of the most interesting and important applications. In this article, a novel Gaussian mixture model (GMM)-based anomaly detection (GMMD) method for HSI is proposed. The main contributions of this article are a new GMM-based extraction approach for extracting the anomaly pixels and an effective GMM-based weighting approach for fusing the extracted anomaly results. Specifically, based on the fact that the spectral values of anomaly pixels in some bands are different from those of background pixels, we propose a GMM-based anomaly extraction approach in which the HSI is characterized by the GMM and the anomaly pixels are extracted by a range prescribed by the GMM parameters. In order to fuse the extracted anomaly results, the GMM-based weighting method is introduced to adaptively construct the detection map. The detection map is rectified by using a guided filter to obtain the final anomaly detection map. Experimental results conducted on four hyperspectral data sets demonstrate the superior performance of the proposed GMMD method.**

***Index Terms*— Anomaly detection, anomaly extraction, band fusion, Gaussian mixture model (GMM), hyperspectral image (HSI).**

I. Introduction

HYPERSPECTRAL remote sensing image has high spectral resolution and plentiful bands, which provides subtle spectral differences between different features. Because of these characteristics, hyperspectral remote sensing image is widely used in many fields, e.g., ecological monitoring [1], mineral exploration [2], environmental monitoring [3], [4], and land cover classification [5], [6]. With the development of hyperspectral remote sensing, various hyperspectral imagery processing has also been developed, such as hyperspectral image (HSI) fusion [7], [8], super-resolution [9], target detection [10], anomaly detection [11], clustering [12], [13], and classification [14], [15]. Among them, HSI detection has received increasing interest and attention due to the needs of applications. The HSI target detection algorithms are based on some prior information, but it is difficult to obtain the prior information in practical applications. Anomaly detection can detect targets that have spectral differences from the surrounding environment without prior information [16]. HSI anomaly detection has practical significance for various applications. Anomaly detection can provide the preliminary results and possible locations of interested regions for target detection and identification. It can also find the possible locations of the targets, eliminate most of background data, and reduce the false-positive rate.

In recent years, a number of HSI anomaly detection methods have been proposed. Among them, many methods use different ways to model the background. The Reed–Xiaoli (RX) method proposed by Reed and Yu [17], an important milestone of anomaly detection, is a typical representative of this kind. The RX method utilizes the multivariate Gaussian distribution to characterize the background and exploits the Mahalanobis distance between the test pixel and the background to detect anomalies. The RX detector has two typical versions: global RX and local RX [18]. The global RX calculates the mean and covariance matrix of background from the full image, whereas in the local RX algorithm, the background is estimated from the local neighborhood around the test pixel. In addition, a number of improved algorithms based on the RX method have been proposed [19]–[23]. For example, Kwon and Nasrabadi [19] proposed a kernel RX algorithm that is the nonlinear variant of the RX and transforms the HSI into a high-dimensional feature space. The weighted-RX method [21] estimates the background information by reducing the weight of the anomaly pixels and increasing the weight of the background.

The above algorithms that adopt different ways to extract background knowledge are statistical approaches. Besides these statistical approaches, the representation-based methods

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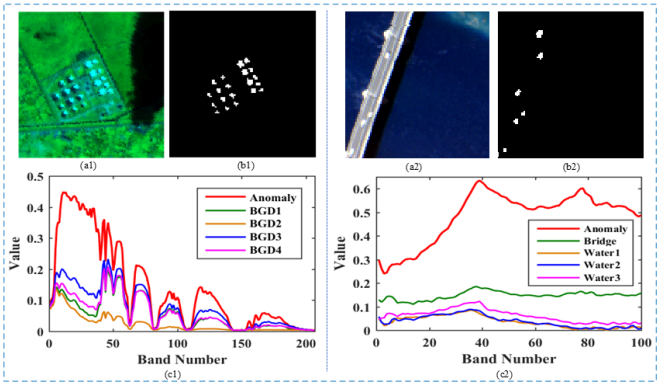


Fig. 1. (a1) and (a2) Hyperspectral images. (b1) and (b2) Anomaly location (ground-truth map). (c1) and (c2) Spectral signatures of the anomaly and background (BGD).

are another class for HSI anomaly detection [24]–[26]. The collaborative representation-based algorithm (CRD) [24] adaptively estimates the background and uses the notion that the pixels of backgrounds are approximately represented by the spatial neighborhoods [27]. Then, the anomalies that cannot be represented are detected from the residual image [24]. In [25], the orthogonal background subspace is adaptively estimated by the background joint sparse representation, and the anomalies are highlighted by an unsupervised adaptive subspace detection method.

Apart from the aforementioned methods, the low-rank-based methods have emerged for anomaly detection recently. The robust principal component analysis (PCA)-based anomaly algorithm assumes that the background is low rank. It projects the HSI into a low-rank component that includes background and a sparse component that includes anomalies [28]. In [29], the low-rank and sparse representation-based anomaly method extracts anomaly pixels from the residual between the original HSI and the background used in the low-rank representation. Moreover, the low-rank and sparse matrix decomposition (LRaSMD)-based algorithm also assumes that the background images are low rank and separates the sparse anomalies from the low-rank background by solving a constrained convex optimization problem [30].

The statistical approaches cannot be able to effectively remove background information because the background information may not obey Gaussian distribution, especially in the case of HSI with complex and variable feature categories. When the constructed dictionary is not sufficiently discriminative, the representation-based methods may also not be effective for eliminating background interference. In order to effectively remove background information and reduce the interference of background information, a novel Gaussian mixture model-based anomaly detection (GMMD) method is proposed in this article. Gaussian mixture model (GMM) uses a linear combination of two or more Gaussian distributions to characterize the data. These multiple Gaussian distributions are mixed in varying proportions. Fig. 1 shows the spectral curves of the anomaly and background of two hyperspectral data. It can be seen from Fig. 1 that the anomaly pixels have abnormal spectral signatures that are different from the spectral curves of the background pixels. As shown in Fig. 1, we can also see that the spectral values of the anomaly pixels are abnormal in most bands compared with the spectral values of the background pixels in the corresponding bands. In other words, the anomalies have abnormal spectral vectors compared with surrounding background, and their spectral values in some bands are different from the spectral values of background pixels. Based on this fact, we utilize the GMM to characterize each band of HSI and extract anomaly pixels of each band via using a prescribed range. This GMM-based extraction method effectively suppresses background information. After the anomaly pixels of each band are extracted, a GMM-based weighting method is proposed to construct the anomaly detection map, and this constructed detection map is rectified by guided filtering. Furthermore, before anomaly detection, the dimension of HSI is reduced by band partitioning and grouping. In Section IV, we analyze the effectiveness of dimension reduction preprocessing on each competitive method and the proposed GMMD method. Comparative analyses validate that the dimension reduction algorithms described in this article greatly decrease the computing time and improve the detection performance of the GMMD method. The proposed GMMD method is evaluated using one simulated data set and three realistic data sets and has a better performance compared with other methods.

The remainder of this article is organized as follows. In Section II, the GMM is briefly reviewed. Section III gives a detailed description of the proposed GMMD method. The experimental results and discussions are presented in Section IV. Finally, conclusions are drawn in Section V.

II. GMM

GMM is a mixed model composed of K single Gaussian models [31], [32]. It uses a linear combination of multiple Gaussian distributions to characterize the data distribution. Each single Gaussian distribution is called a component of GMM. The number of components is considered as the number of categories. The probability density function of the GMM is defined as

$$P(x) = \sum_{k=1}^{K} \alpha_k \mathcal{N}(x|\mu_k, \theta_k) \quad \text{s.t.} \sum_{k=1}^{K} \alpha_k = 1 \tag{1}$$

where K is the number of mixture components, $\alpha_k \geq 0$ is the proportion of the kth Gaussian distribution in GMM, $\mathcal{N}(x|\mu_k, \theta_k)$ represents the kth Gaussian distribution, and μ_k and θ_k are the mean and variance, respectively.

The distribution of data can be approximated by the GMM. The parameters α_k, μ_k, and θ_k can be calculated by maximizing the log-likelihood function. The expectation–maximization (EM) algorithm [33], [34] is an effective method of maximizing the likelihood function and can be utilized to estimate the parameters of GMM. The GMM has been widely applied in various image processing, including clustering [35], classification [36], [37], segmentation [38], [39], and registration [40]. In this article, the GMM is applied on HSI anomaly detection, and an effective and new GMM-based anomaly detection algorithm is proposed, which is described in detail in Section III.

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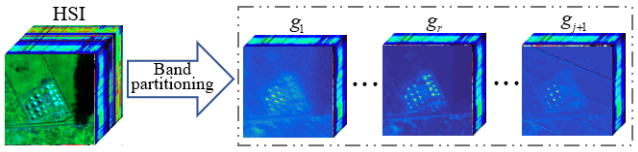


Fig. 2. Schematic of band partitioning.

III. Proposed Method

The main process of the GMMD algorithm is characterized in Sections III-A–III-D. The HSI is partitioned into some subsets of bands by analyzing the correlation between adjacent bands, and then, the partitioned bands are fused to accomplish HSI dimension reduction. After that, we propose a GMM-based anomaly extraction approach to extract the anomaly pixels of each band and construct the anomaly detection map by using a GMM-based weighting method.

A. Band Partitioning

To improve computational efficiency and detection performance, the dimension of HSI can be reduced due to high spectral correlation. The closer the bands in an HSI, the stronger the correlation. The cross correlation (CC) between the adjacent bands is utilized to partition the HSI. Larger CC indicates stronger correlation. The CC between two adjacent bands is defined as

$$\mathrm{CC}(\mathbf{X}^i, \mathbf{X}^{i+1}) = \frac{\sum_{m=1}^{M}\sum_{n=1}^{N}\left(\mathbf{X}_{m,n}^{i}-\beta^i\right)\left(\mathbf{X}_{m,n}^{i+1}-\beta^{i+1}\right)}{\sqrt{\sum_{m=1}^{M}\sum_{n=1}^{N}\left(\mathbf{X}_{m,n}^{i}-\beta^i\right)^2}\sqrt{\sum_{m=1}^{M}\sum_{n=1}^{N}\left(\mathbf{X}_{m,n}^{i+1}-\beta^{i+1}\right)^2}} \tag{2}$$

where $\mathbf{X} \in \mathrm{R}^{M\times N\times\tau}$ denotes the HSI, M and N represent the number of rows and columns of HSI, τ is the number of the HSI bands, $\mathbf{X}^i$ is the ith band of HSI, and β^i is the mean of the ith HSI band. The highly correlated bands should be divided into the same subspace, while the bands with weak correlation are divided into different subspaces. Thus, we find the local minimum values of CC. When the CC value at a point is smaller than the CC values at both sides of this point, this point is considered to be the local minimum point. Let $[s_{l_1}, s_{l_2}, \ldots, s_{l_j}]$ represent the local minimum values of CC and $[l_1, l_2, \ldots, l_j]$ represent the corresponding local minimum points. These local minimum points are served as the segmentation points to partition the HSI. For instance, l_1 is a local minimum point and is served as the segmentation point. The CC value between bands $\mathbf{X}^{l_1}$ and $\mathbf{X}^{l_1+1}$ is smaller than the CC values at both sides of l_1. Thus, $\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^{l_1}$ is divided into the first group of bands, and $\mathbf{X}^{l_1+1}, \mathbf{X}^{l_1+2}, \ldots, \mathbf{X}^{l_2}$ is divided into the second group of bands. As shown in Fig. 2, the HSI is partitioned as the following equation:

$$\mathbf{X} = (g_1, g_2, \ldots, g_r, \ldots, g_{j+1}) \tag{3}$$

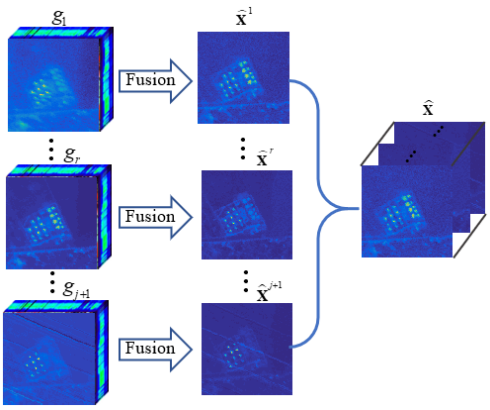


Fig. 3. Schematic of partitioned bands' fusion.

$$\begin{cases} g_1 = (\mathbf{X}^1, \mathbf{X}^2, \ldots, \mathbf{X}^{l_1}) \\ \ldots \\ g_r = (\mathbf{X}^{l_{r-1}+1}, \mathbf{X}^{l_{r-1}+2}, \ldots, \mathbf{X}^{l_r}) \\ \ldots \\ g_{j+1} = (\mathbf{X}^{l_j+1}, \ldots, \mathbf{X}^{\tau}) \end{cases} \tag{4}$$

where $(g_1, g_2, \ldots, g_r, \ldots, g_{j+1})$ denotes $j+1$ groups of bands and g_r represents the rth group.

B. Partitioned Bands' Fusion

To fuse the partitioned bands in each group, three fusion algorithms are adopted as follows. Let $\widehat{\mathbf{X}} \in \mathrm{R}^{M\times N\times(j+1)}$ refer to the dimension reduced HSI and $\widehat{\mathbf{X}}^r$ denote the rth band of $\widehat{\mathbf{X}}$. $\widehat{\mathbf{X}}^r$ is also the fused image of the rth group of bands. Fig. 3 shows the schematic of the partitioned bands fusion.

The first strategy is an adaptive weighted (ADW) algorithm. The neighboring bands in one partitioned group are highly correlated and have similar information. To preserve most of useful information, for the rth group, the PCA transform is applied to the rth group to obtain the first principal component $\mathbf{P}_1^r$, and the fused image $\widehat{\mathbf{X}}^r$ is achieved through minimizing the difference between $\widehat{\mathbf{X}}^r$ and $\mathbf{P}_1^r$. This procedure is described as the following optimization equation:

$$\min \ \|\widehat{\mathbf{X}}^r - \mathbf{P}_1^r\|^2 = \min_{\omega_i} \left\| \sum_{i=l_{r-1}+1}^{l_r} \omega_i \mathbf{X}^i - \mathbf{P}_1^r \right\|^2, \quad \omega_i \geq 0 \tag{5}$$

for $r = 1, 2, \ldots, j+1$, where $\mathbf{P}_1^r$ is the first principal component of the rth group and $\{\omega_i\}_{i=l_{r-1}+1,\ldots,l_r}$ are the weights. The constraint term ensures that the weights are nonnegative. The Lagrangian multiplier is utilized to offer an equivalent formulation to (5)

$$E(\omega_i, \eta_i) = \left\| \sum_{i=l_{r-1}+1}^{l_r} \omega_i \mathbf{X}^i - \mathbf{P}_1^r \right\| + \eta_i(-\omega_i) \tag{6}$$

where $\eta \geq 0$ is the Lagrangian multiplier. We use the gradient descent method to minimize $E(\omega_i, \eta_i)$. The rth band of the dimension reduced HSI is subsequently obtained by computing $\widehat{\mathbf{X}}^r = \sum_{i=l_{r-1}+1}^{l_r} \omega_i \mathbf{X}^i$.

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The PCA algorithm, averaging weighted (AVW) algorithm, and the taking the largest absolute value (TLAV) algorithm are the remaining three alternative methods. They are three classical fusion methods. For the PCA algorithm, the first principal component of the rth group $\mathbf{P}_1^r$ is obtained by using the PCA transform, and $\mathbf{P}_1^r$ is extracted as the fusion band $\widehat{\mathbf{X}}^r$. For the TLAV algorithm, the pixel value with the largest absolute value in partitioned band is extracted as the value of the corresponding pixel in the fusion band. For the AVW algorithm, $\widehat{\mathbf{X}}^r$ is calculated as

$$\widehat{\mathbf{X}}^r = \sum_{i=l_{r-1}+1}^{i=l_r} \frac{\mathbf{X}^i}{l_r - l_{r-1}} \tag{7}$$

for $r = 1, 2, \ldots, j+1$. These three algorithms are all simple and effective fusion methods. The proposed ADW algorithm adaptively obtains the weights and retains most useful information. The PCA algorithm also obtains most important information. The AVW algorithm is fast and has good image structural integrity. The TLAV algorithm has the advantages of high edge strength and clear texture. We analyze the performance of these four bands fusion algorithms in Section IV.

C. Anomaly Extraction

After obtaining the dimension reduced HSI $\widehat{\mathbf{X}}$, a GMM-based anomaly extraction method is proposed in this step. The GMM can approximately characterize the distribution of the data, and thus, we assume that the distribution of $\widehat{\mathbf{X}}^r$ obeys the GMM. The density function of $\widehat{\mathbf{X}}^r$ is expressed as

$$\begin{aligned} p\left(\widehat{\mathbf{X}}_d^r\right) &= \sum_{t=1}^{T^r} \alpha_t^r \mathcal{N}\left(\widehat{\mathbf{X}}_d^r \middle| \mu_t^r, \theta_t^r\right) \\ \text{s.t.} \sum_{t=1}^{T^r} \alpha_t^r &= 1, \quad r = 1, 2, \ldots, j+1 \end{aligned} \tag{8}$$

where $\widehat{\mathbf{X}}^r$ denotes the rth band of $\widehat{\mathbf{X}}$, $d = 1, 2, \ldots, MN$, MN denotes the number of pixels in one HSI band, $\widehat{\mathbf{X}}_d^r$ is the dth pixel of $\widehat{\mathbf{X}}^r$, T^r is the number of mixture components of the rth band, $\alpha_t^r \geq 0$ is the weight of the tth Gaussian component of $\widehat{\mathbf{X}}^r$, and μ_t^r and θ_t^r represent the mean and covariance of the tth Gaussian component of $\widehat{\mathbf{X}}^r$, respectively.

Given a dimension reduced HSI $\widehat{\mathbf{X}}$, the rth band $\widehat{\mathbf{X}}^r$ follows the GMM distribution, and the parameters $\{\alpha_t^r, \mu_t^r, \theta_t^r\}_{t=1,2,\ldots,T^r, r=1,2,\ldots,j+1}$ can be estimated by the EM algorithm. In the EM algorithm, the likelihood function of $\widehat{\mathbf{X}}^r$ is first obtained. Then, the E step calculates the posterior probability of the tth Gaussian component, and the M step combines the above posterior probability to maximize the likelihood function of $\widehat{\mathbf{X}}^r$ to calculate the parameters. The GMM parameters can be formulated using the following equation:

$$\begin{aligned} \alpha_t^r &= \frac{\sum_{d=1}^{MN} \gamma_{t,d}^r}{MN} \\ \mu_t^r &= \frac{\sum_{d=1}^{MN} \gamma_{t,d}^r \widehat{\mathbf{X}}_d^r}{\sum_{d=1}^{MN} \gamma_{t,d}^r} \\ \theta_t^r &= \frac{\sum_{d=1}^{MN} \gamma_{t,d}^r \left(\widehat{\mathbf{X}}_d^r - \mu_t^r\right)^2}{\sum_{d=1}^{MN} \gamma_{t,d}^r} \end{aligned} \tag{9}$$

where $\gamma_{t,d}^r = ((\alpha_t^r \mathcal{N}(\widehat{\mathbf{X}}_d^r | \mu_{t_r}, \theta_{t_r}))/(\sum_{t=1}^{T^r} \alpha_t^r \mathcal{N}(\widehat{\mathbf{X}}_d^r | \mu_t^r, \theta_t^r)))$ is the posterior probability at the dth pixel of the tth component of $\widehat{\mathbf{X}}^r$. Before using the EM algorithm, the posterior probability is first generated randomly, and the rows of posterior probability add up to 1. The parameters $\{\alpha_t^r, \mu_t^r, \theta_t^r\}_{t=1,2,\ldots,T^r, r=1,2,\ldots,j+1}$ are initialized by the generated posterior probability and (9). The E step and M step are repeated until the likelihood function of $\widehat{\mathbf{X}}^r$ converges.

Among the parameters $\{\alpha_t^r, \mu_t^r, \theta_t^r\}_{t=1,2,\ldots,T^r, r=1,2,\ldots,j+1}$, α_t^r and μ_t^r represent the proportion and mean of the tth Gaussian component for the rth band of $\widehat{\mathbf{X}}$, respectively. When $\widehat{\mathbf{X}}^r$ follows a GMM distribution shown in (8), $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ represents the mean of $\widehat{\mathbf{X}}^r$. Let $\rho^r = ((\sum_{d=1}^{MN} \widehat{\mathbf{X}}_d^r)/MN)$ denote the arithmetic mean of $\widehat{\mathbf{X}}^r$. Instead of using the arithmetic mean $\rho^r = ((\sum_{d=1}^{MN} \widehat{\mathbf{X}}_d^r)/MN)$, we utilize $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ to represent the mean of $\widehat{\mathbf{X}}^r$ based on the principle of GMM. Since the distribution of $\widehat{\mathbf{X}}^r$ obeys the GMM, it is valid and reasonable to use $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ to represent the mean of $\widehat{\mathbf{X}}^r$.

Anomaly targets have spectral vectors that are different from the surrounding environment. The spectral values of anomaly pixels in some bands are abnormal compared with those of background pixels. The anomaly pixels in $\widehat{\mathbf{X}}^r$ are extracted by using the following equation:

$$\mathbf{D}_d^r = \begin{cases} \widehat{\mathbf{X}}_d^r, & \text{if } \widehat{\mathbf{X}}_d^r > \sum_{t=1}^{T^r} \alpha_t^r \mu_t^r + 2\sigma^r \\ \widehat{\mathbf{X}}_d^r, & \text{if } \widehat{\mathbf{X}}_d^r < \sum_{t=1}^{T^r} \alpha_t^r \mu_t^r - 2\sigma^r \\ 0, & \text{otherwise} \end{cases} \tag{10}$$

for $r = 1, 2, \ldots, j+1$, $d = 1, 2, \ldots, MN$, where $\mathbf{D}^r$ is the anomaly extraction result of $\widehat{\mathbf{X}}^r$, $\mathbf{D}_d^r$ is the dth pixel of $\mathbf{D}^r$, and σ^r denotes the standard deviation of $\widehat{\mathbf{X}}^r$. The spectral values of most pixels are in the range $[\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r - 2\sigma^r, \sum_{t=1}^{T^r} \alpha_t^r \mu_t^r + 2\sigma^r]$, and the pixels that are not in this range are considered as anomaly pixels. Thus, as shown in (10), the values of these anomaly pixels are extracted and retained. Fig. 4 shows this anomaly extraction step. From Fig. 4, it can be seen that the anomaly pixels of each band are effectively extracted.

D. Detection Map Construction

The anomaly detection map is constructed by using a novel GMM-based weighting method. Based on the concept of GMM, GMM characterizes the data by using a linear combination of two or more Gaussian distributions, and thus, the distribution of $\mathbf{D}^r$ obeys the GMM. Furthermore, from (10), it can be seen that $\mathbf{D}^r$ is a part of $\widehat{\mathbf{X}}^r$. $\widehat{\mathbf{X}}^r$ is the linear transformation of $\mathbf{X}^i$, so the values of $\mathbf{D}^r$ that are not zero are also the linear transformation of the corresponding values of $\mathbf{X}^i$. Based on the property of the Gaussian distribution described in the previous subsection, the nonzero data of $\mathbf{D}^r$

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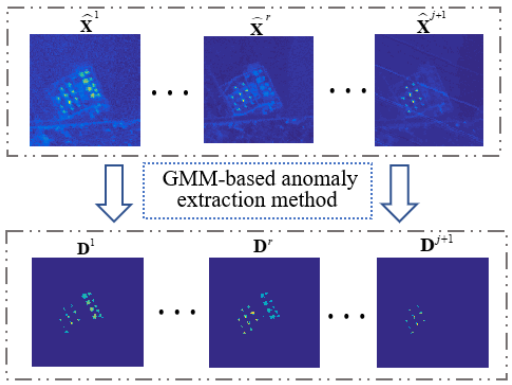


Fig. 4. Schematic of the GMM-based anomaly extraction.

follow GMM. For the nonzero values of $\mathbf{D}^r$, the density function is expressed as

$$P(\mathbf{D}_e^r) = \sum_{h=1}^{H^r} \widehat{\alpha}_h^r \mathcal{N}(\mathbf{D}_e^r | \widehat{\mu}_h^r, \widehat{\theta}_h^r)$$
$$\text{s.t.} \sum_{h=1}^{H^r} \widehat{\alpha}_h^r = 1, \quad r = 1, 2, \ldots, j+1 \tag{11}$$

where $e = 1, 2, \ldots, E^r$, E^r denotes the number of the nonzero values of $\mathbf{D}^r$, $\mathbf{D}^r$ is the rth band of $\mathbf{D}$, $\mathbf{D}_e^r$ is the eth pixel of the nonzero values of $\mathbf{D}^r$, H_r is the number of Gaussian components of $\mathbf{D}^r$, and $\widehat{\alpha}_h^r \geq 0$, $\widehat{\mu}_h^r$, and $\widehat{\theta}_h^r$ represent the weight, mean, and covariance of the hth Gaussian component of $\mathbf{D}^r$, respectively. The parameters $\{\widehat{\alpha}_h^r, \widehat{\mu}_h^r, \widehat{\theta}_h^r\}_{h=1,2,\ldots,H^r, r=1,2,\ldots,j+1}$ can be estimated by the EM algorithm. $\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r$, which is the mean of $\mathbf{D}^r$, can represent the intensity of $\mathbf{D}^r$. Larger $\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r$ means more extracted anomaly pixels in the rth band. Therefore, this band should take a larger proportion. The anomaly detection map is obtained by the following weighted formula:

$$\widehat{\mathbf{S}} = \sum_{r=1}^{j+1} \left(\frac{\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r}{\sum_{r=1}^{j+1} \sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r} \mathbf{D}^r \right) \tag{12}$$

where $\widehat{\mathbf{S}}$ is the anomaly detection map.

To make the adjacent pixels belonging to the same object have similar values and avoid artifacts at the boundaries, the anomaly detection map $\widehat{\mathbf{S}}$ is rectified by guided filtering. The guided filter [41], [42] smooths the input image while preserving the edge. $\widehat{\mathbf{S}}$ is the input image, and the first band of dimension reduced HSI $\widehat{\mathbf{X}}^1$ is served as the guidance image. The filtered image $\mathbf{S}$ is the local linear transformation of the guidance image $\widehat{\mathbf{X}}^1$, and the difference between $\mathbf{S}$ and $\widehat{\mathbf{S}}$ should be minimized

$$\mathbf{S}_d = b_q \widehat{\mathbf{X}}_d^1 + c_q \quad \forall d \in \Lambda_q \tag{13}$$
$$\min \sum_{d \in \Lambda_q} \left(\left(b_q \widehat{\mathbf{X}}_d^1 + c_q - \widehat{\mathbf{S}}_d \right)^2 + \lambda b_q^2 \right) \tag{14}$$

for $q = 1, 2, \ldots, MN$, where the output image $\mathbf{S}$ represents the final anomaly detection map, $\mathbf{S}_d$ represents the dth pixel of $\mathbf{S}$, Λ_d is a square window centered at the pixel d, b_q and c_q are the linear coefficients, and λ is a regularization parameter and is set to 10^{-6}. The linear coefficients b_q and c_q are calculated by solving (17)

$$b_q = \left(\chi_q^2 + \lambda \right)^{-1} \left(\frac{1}{|\Lambda|} \sum_{d \in \Lambda_q} \widehat{\mathbf{X}}_d^1 \widehat{\mathbf{S}}_d - \delta_q \widetilde{s}_q \right)$$
$$c_q = \widetilde{s}_q - b_q \delta_q \tag{15}$$

where $|\Lambda|$ is the number of pixels in Λ_d, δ_q, and χ_q^2 denote the mean and variance of $\widehat{\mathbf{X}}^1$ in Λ_q, and $\widetilde{s}_q$ is the mean of $\widehat{\mathbf{S}}$ in Λ_q. The pixel d is involved in all the overlapping windows Λ_q that covers d. Thus, the filtering output is obtained after computing the linear coefficients b_q and c_q for all windows Λ_q. An example of this procedure is shown in Fig. 5. The anomaly detection map $\widehat{\mathbf{S}}$ is adaptively constructed by the GMM-based weighting method. It can be observed that the guided filter can effectively adjust the detection map such that the neighboring pixels of the same object tend to have similar values.

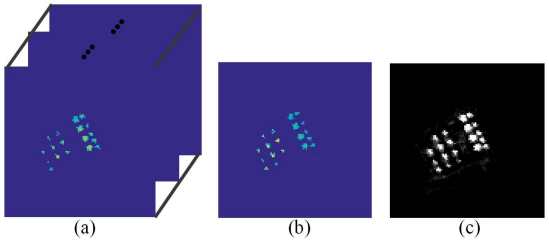


Fig. 5. Detection map construction. (a) Anomaly extraction result $\mathbf{D}$. (b) Anomaly detection map $\widehat{\mathbf{S}}$. (c) Rectified final anomaly detection map $\mathbf{S}$.

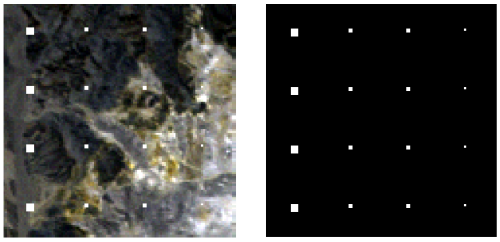


Fig. 6. (a) Simulated image in RGB pseudocolor image. (b) Ground-truth map.

IV. Experimental Results and Discussion

A. Data Set Description

To evaluate the performance of the proposed method, experiments are conducted on one simulated hyperspectral data set and three real hyperspectral data sets. The first data set is the simulated data whose background information was collected by the Airborne Visible/Infrared Imaging Spectrometer. The background information of this experimental data is a mining area in Nevada, USA. The simulated hyperspectral data are synthesized from the ground-truth spectral information of the real data. These data contain 120×120 pixels and 224 bands. The noise bands and the water vapor absorption bands

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TABLE I
FOUR MIXED PIXELS IN THE THIRD COLUMN

Row	The generated ways of four mixed pixels	
1	$0.5A + 0.5B$	$0.5A + 0.5C$
	$0.5A + 0.5K$	$0.5A + 0.5M$
2	$0.5A + 0.5B$	$0.5B + 0.5C$
	$0.5B + 0.5K$	$0.5B + 0.5M$
3	$0.5A + 0.5C$	$0.5B + 0.5C$
	$0.5C + 0.5K$	$0.5C + 0.5M$
4	$0.5A + 0.5K$	$0.5B + 0.5K$
	$0.5C + 0.5K$	$0.5K + 0.5M$

TABLE II
FOUR SUBPIXELS IN THE FOURTH COLUMN

Row	The generated ways of four mixed pixels
1	$0.5A + 0.5b$
2	$0.5B + 0.5b$
3	$0.5C + 0.5b$
4	$0.5K + 0.5b$

are removed, and 188 bands are used for the experiments. In order to generate the synthetic data, the anomaly targets are implanted in the background data, and the white noise with an SNR of 20:1 is also added. The RGB pseudocolor image of the synthetic data set is shown in Fig. 6(a), and these data have 16 anomaly targets. Five mineral signatures that are A (alunite), B (buddingtonite), C (calcite), K (kaolinite), and M (muscovite) and a background signature b are utilized to generate the anomaly targets. Among the 16 anomaly targets, each column has four anomaly targets that have the same size and different categories. The first column contains four 4×4 pure pixels that are A, B, C, and K mineral signatures, respectively. The second column is four 2×2 pure pixels that are also A, B, C, and K mineral signatures. There are four 2×2 mixed pixels in the third column and four 1×1 subpixels in the fourth column. The generated ways of four mixed pixels and the generated ways of four subpixels are shown in Tables I and II, respectively. The ground truth is shown in Fig. 6(b).

The second HSI shown in Fig. 7(a) was collected by the Reflective Optics System Imaging Spectrometer sensor [43]. This HSI is a real hyperspectral scene of the city center of Pavia in Northern Italy, and the RGB pseudocolor image and the ground-truth map are shown in Fig. 7(a) and (d), respectively. The Pavia image scene has a size of 100×100; 102 bands spanning the wavelength interval of $0.43 - 0.86$ μm are used for experiments, and the spatial resolution is 1.3 m. This data set covers bridge and water that are the main background materials, and some vehicles on the bridge and bare soil near the pier which are the anomaly target pixels. The percentage of anomaly target pixels in the total HSI is 0.68%.

The data set shown in Fig. 7(b) was provided from the Airport-Beach-Urban data set that was captured by the

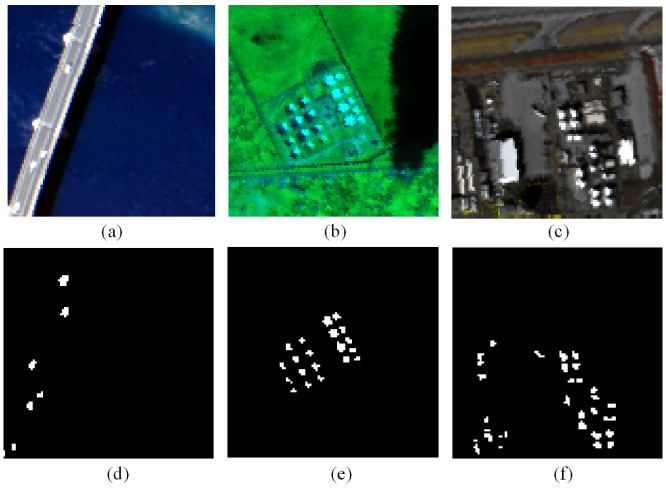


Fig. 7. (a) Pavia image scene in RGB pseudocolor image. (b) Texas Coast image scene in RGB pseudocolor image. (c) Los Angeles image scene in RGB pseudocolor image. (d)–(f) Their corresponding ground-truth maps.

Airborne Visible/Infrared Imaging Spectrometer. This data set is a scene of Texas Coast and was collected on August 29, 2010 [44]. The Texas Coast image shown in Fig. 7(b) contains 100×100 pixels, and the spatial resolution is 17.2 m. After removing the noisy bands in the Texas Coast HSI, 207 bands are used for experiments. The ground-truth map of the Texas Coast image is shown in Fig. 7(e), and the percentage of anomaly pixels is 1.55%.

The fourth data set used in the experiments was collected at the location of the Los Angeles City and was acquired by the Airborne Visible/Infrared Imaging Spectrometer [44]. This data set contains 100×100 pixels, as shown in Fig. 7(c). Fig. 7(f) shows the ground-truth map. The spatial resolution of the Los Angeles image is 7.1 m per pixel. Each pixel contains 205 dimensions, in which noisy bands have been removed.

B. Detection Performance Analysis of Dimension Reduction Preprocessing

To evaluate the performance of the proposed GMMD method, the receiver operating characteristic (ROC) curve and the area under the ROC curve (AUC) are adopted [45], [46]. For the ROC curve, at the same false alarm rate (FAR), a higher detection rate indicates a better detection result. The AUC is obtained by computing the whole AUC. A detection method that has the larger AUC value shows better performance. The detection capability of the proposed GMMD method is assessed through comparison with several superior methods, which are global RX (GRX), local RX (LRX), CRD, and LRaSMD. GRX and LRX are two typical versions of the RX, which is based on the Mahalanobis distance. The GRX and LRX estimate the background statistics of an entire image and a local area, respectively. CRD and LRaSMD are two state-of-the-art algorithms. CRD is a collaborative representation-based anomaly detection algorithm. LRaSMD that is based on the low-rank and sparse decomposition detects the anomalies by solving a constrained convex optimization problem.

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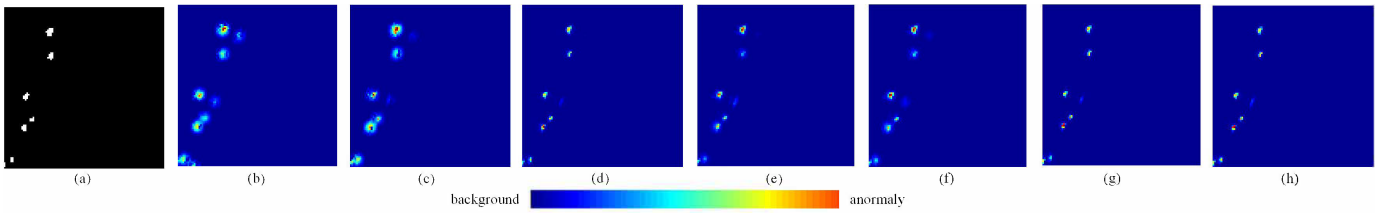


Fig. 8. 2-D plots of detection results of each dimension reduction method in the proposed GMMD method for Pavia image scene. (a) Ground-truth map. (b) H_noDR. (c) ADW. (d) PCA. (e) AVW. (f) TLAV. (g) IES. (h) VAR.

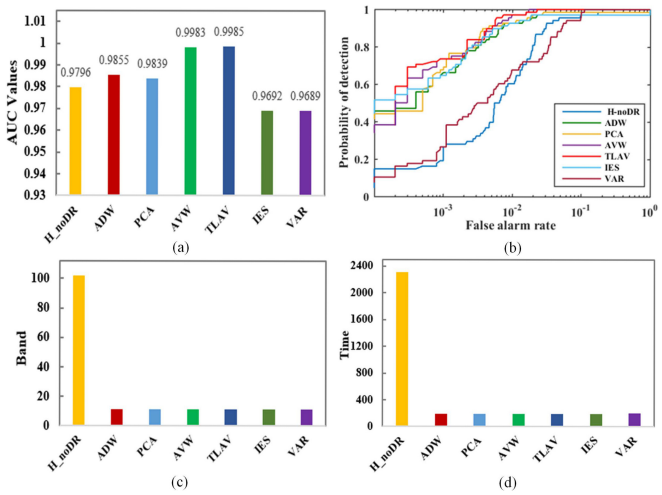


Fig. 9. Detection results of dimension reduction preprocessing in the proposed GMMD method for Pavia image scene. (a) AUC values. (b) ROC curves. (c) Band number. (d) Computing time (seconds).

In the proposed method, we first reduce the dimension of the data sets. The HSI is adaptively divided into subsets according to spectral correlation, and the partitioned bands are combined with the fusion algorithm. To analyze the effectiveness of dimension reduction preprocessing, we compare the detection results of the HSI without dimensionality reduction (H_noDR) and these of the dimensionality reduced HSI. The partitioned bands' fusion step of the proposed method describes three-band fusion algorithms, which are the ADW algorithm, the PCA algorithm, the AVW algorithm, and the TLAV algorithm. We compare these four algorithms and also compare two classical band selection algorithms. The two band selection algorithms that are applied to each group to select one band are the information entropy-based algorithm (IES) and the variance-based algorithm (VAR). The experiments in this article were all performed using MATLAB R2015b. We used the library function (princomp function) to extract the first principal component $\mathbf{P}_1^r$ in the PCA algorithm. In the PCA method, the principal components are derived by taking the eigenvectors of the covariance matrix. The first principal component corresponds to the largest eigenvalue. $\mathbf{P}_1^r$ and $-\mathbf{P}_1^r$ can both be used as the first principal component. $\mathbf{P}_1^r$ is selected as the first principal component in this article.

These algorithms are applied to the Pavia data set, and the detection results are shown in Figs. 8 and 9. As shown in Fig. 9(c), the original HSI has 102 bands, and the dimensionality reduction HSI has 11 bands. The computing times of detection processing after dimension reduction are all greatly reduced, as shown in Fig. 9(d). The 2-D plots of the detection results of each algorithm are shown in Fig. 8(b)–(g). H_noDR and ADW show high values for the anomaly pixels. However, they also show high values for pixels around the anomaly and some bridge pixels, which are not abnormal, and cause false alarms. Compared with H_noDR and ADW, the other five algorithms have lesser FARs. However, the anomaly pixels detected by the PCA, IES, and VAR algorithms are incomplete, and some anomaly pixels are not detected. In contrast, the detection performance of AVW and TLAV is superior. The AUC values and the ROC curves of each algorithm are shown in Fig. 9(a) and (b), respectively. As shown in Fig. 9(a), TLAV yields the largest AUC value, followed by AVW. ADW, PCA, AVW, and TLAV that perform dimensionality reduction processing on HSI have higher values than the H_noDR method. As shown in the ROC curves obtained by the IES and VAR band selection dimension reduction algorithms, when the FAR starts from about 0.01, the probability of detection is almost not increase. The ROC curves of AVW and TLAV show better performance than other methods, and TLAV is the best on the whole. These results in Figs. 8 and 9 indicate that, with the computing times being greatly shortened, the band fusion reduction algorithms (ADW, PCA, AVW, and TLAV) have better detection performance than the H_noDR method, which does not reduce the dimension on HSI, and TLAV works the best. Therefore, in the proposed GMMD method, it is meaningful and effective to reduce the dimensionality of the data sets, and the TLAV algorithm is selected.

In order to analyze the effects of dimension reduction preprocessing of the competitive methods, we test the performance of each comparison method with and without dimension reduction. The AUC scores of each comparison method for three real data sets are shown in Fig. 10. We select the TLAV band fusion algorithm to achieve dimension reduction for each method. The GRX, LRX, CRD, and LRaSMD methods with dimension reduction are named GRX_DR, LRX_DR, CRD_DR, and LRaSMD_DR, respectively. The proposed GMMD with and without dimension reduction is referred to as GMMD_DR and GMMD_DNR, respectively. As shown in Fig. 10, for three data sets, the AUC values of LRX and CRD are greater than those of LRX_DR and CRD_DR, which demonstrates that LRX and CRD are not suitable for dimension reduction preprocessing. The AUC values of GMMD_NDR are lower than the values of GMMD_DR, and

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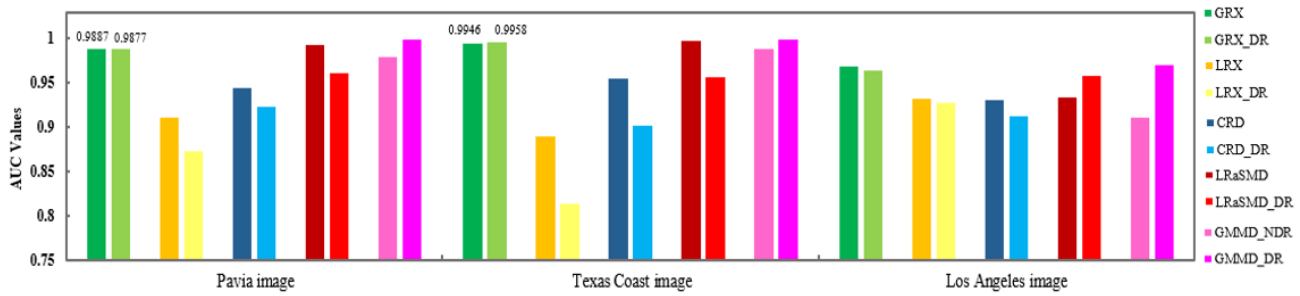


Fig. 10. Detection results of dimension reduction preprocessing for each competitive method.

these results indicate that it is necessary for the proposed GMMD to perform dimensionality reduction. From Fig. 10, it can be seen that the detection results of GRX are superior to GRX_DR on the Pavia and Los Angeles images, but for the Texas Coast image scene, the AUC value of GRX is smaller than the value of GRX_DR. The AUC results of LRaSMD are better than that of LRaSMD_DR for the Pavia and Texas Coast images, while for the Los Angeles data set, the AUC result of LRaSMD_DR is slightly better. These comparisons show that it may be effective to reduce dimension for the LRaSMD and GRX methods as well. For the GMMD method, we propose the GMM-based extraction approach to extract anomaly pixels of each band and introduce the GMM-based weighting method to construct the anomaly detection map. The dimension reduction preprocessing can retain useful bands and information and reduce redundant information. For the GMMD method, dimension reduction is beneficial for improving computational efficiency and detection performance. The comparison methods are based on the spectral vector and the spectral information. The dimension reduction preprocessing may cause the loss of spectral information, which improves calculation efficiency but may reduce detection accuracy. GRX is a global anomaly algorithm, and it may be less affected by the dimension reduction processing. For GRX, dimension reduction processing does not necessarily reduce detection accuracy. Thus, GRX gets improved detection performance for the Texas Coast data and degraded performance for the Pavia and Los Angeles images. LRX and CRD are the local algorithms and may be greatly affected by the dimension reduction processing. The dimension reduction processing generally degrades detection performance for LRX and CRD. Through the above experiments and analysis, in our experiments, we perform dimension reduction preprocessing on the GRX, LRaSMD, and the proposed methods and compare the detection performance of GRX, GRX_DR, LRX, CRD, LRaSMD, LRaSMD_DR, and GMMD.

C. Detection Performance Analysis of Different Ranges of the GMM-Based Anomaly Extraction Method

In the anomaly extraction step of the proposed method, the spectral values of the pixels that are not in the range $[\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r - 2\sigma^r, \sum_{t=1}^{T^r} \alpha_t^r \mu_t^r + 2\sigma^r]$ are extracted and considered as the anomaly pixels. For the sake of analyzing the detection performance of different ranges and verifying the validity of the $\pm 2\sigma^r$ range, the detection results with different ranges are compared. The compared ranges include $\pm 0.5\sigma^r$, $\pm 1\sigma^r$, $\pm 1.5\sigma^r$, $\pm 2\sigma^r$, and $\pm 2.5\sigma^r$. The GMMD method with these different ranges is applied to the Pavia image, and the AUC values and the ROC curves of the GMMD method with different ranges are shown in Fig. 11(a) and (b). The GMMD method with these different ranges is abbreviated as 0.5, 1, 1.5, 2 and 2.5. The AUC values in Fig. 11(a) reveal that the $\pm 2\sigma^r$ range achieves the largest value of 0.9985, and the ROC curves shown in Fig. 11(b) also indicate that the ROC curve of the $\pm 2\sigma^r$ range is closer to the top-left corner than the other ranges. From the above detection results, it can be seen that the GMMD method with the $\pm 2\sigma^r$ range obtains the best performance, and the $\pm 2\sigma^r$ range is selected in this article.

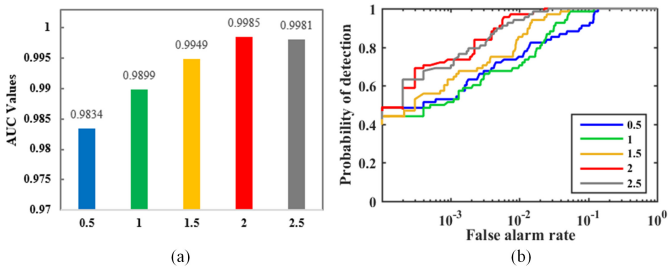


Fig. 11. Detection results of the proposed GMMD method with different ranges for Pavia image scene. (a) AUC values. (b) ROC curves.

D. Parameters Selection and Detection Performance Analysis of the GMM-Based Mean Representation

In the GMMD method, there are two important parameters T^r and H^r. T^r is the number of mixture components for $\widehat{\mathbf{X}}^r$. We adopt the Bayesian information criterion (BIC) criterion [47], [48] to determine T^r. The BIC criterion is a popular model parameter selection method. H^r is the number of mixture components for $\mathbf{D}^r$. According to (10), $\mathbf{D}^r$ includes zero values that are the background pixels and the nonzero values that are considered as the anomaly pixels. From (10), it can be seen that the nonzero values in $\mathbf{D}^r$ include a number of larger values and some smaller values. Therefore, we consider that the nonzero data of $\mathbf{D}^r$ can be approximately characterized by a linear combination of two Gaussian distributions, and H_r is set to 2.

In the GMMD method, $\widehat{\mathbf{X}}^r$ and $\mathbf{D}^r$ follows the GMM distribution. $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ and $\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r$ are the means of

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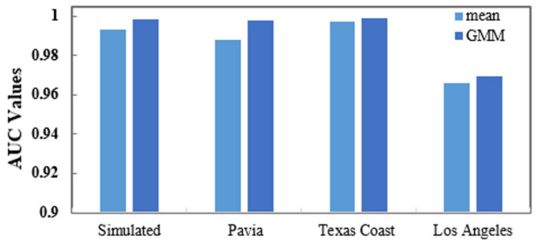


Fig. 12. Detection results of the sample mean-based method and the GMM-based method.

$\widehat{\mathbf{X}}^r$ and $\mathbf{D}^r$, respectively. Since the distribution of $\widehat{\mathbf{X}}^r$ and $\mathbf{D}^r$ obeys the GMM, when the numbers of components T^r and H^r are selected properly, $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ and $\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r$ are valid and reasonable. In order to validate the availability of the GMM-based mean representation, we compare the detection results of the arithmetic mean-based method and the results of the proposed GMM-based mean method. The AUC scores of these two methods for four experimental data sets are shown in Fig. 12. The arithmetic mean-based method and the proposed GMM-based mean method are named mean and GMM, respectively. As shown in Fig. 12, it can be seen that the detection performance of the proposed GMM-based mean method is comparable with the detection performance of the arithmetic mean-based method, and the performance of the proposed GMM-based mean method is slightly better than that of the arithmetic mean-based method. These detection results validate the effectiveness of the proposed GMM-based mean method. These results also show that the selection of parameters is appropriate, and it is valid to use $\sum_{t=1}^{T^r} \alpha_t^r \mu_t^r$ and $\sum_{h=1}^{H^r} \widehat{\alpha}_h^r \widehat{\mu}_h^r$ to represent the mean of $\widehat{\mathbf{X}}^r$ and $\mathbf{D}^r$.

E. Detection Performance Analysis of Guided Filtering Rectification Processing

In the detection map construction step of the proposed method, the detection map is constructed by a GMM-based weighting method and this detection map is rectified by guided filtering to obtain the final detection map. In the rectified procedure, the rectified results are influenced by the window size of the guided filter. In practice, the window size of guided filter is generally selected based on experience. We have carried out a large number of experiments on the guided filter and obtained an empirical result for the selection of the window size. When the guidance image contains more spatial information, the window size of guided filter should be larger. If the guidance image contains less spatial details, the window size should be selected as a smaller value. For the experimental data sets, the guidance images of the Pavia image data and the simulated data contain less spatial details, so the window size of guided filter is a relatively small value for the Pavia image data and the simulated data. Since the guidance images of the Texas Coast data and the Los Angeles data contain more spatial details, the window size of guided filter is a larger value for the Texas Coast data and the Los Angeles data. To analyze the influences of the window size, the GMMD approach is tested on the Pavia image and the Texas Coast data to observe the performance with different window size settings. Fig. 13(a) and (b) shows the detection performance with different window size settings for the Pavia and Texas Coast data, respectively. When the window size is 5, the GMMD method acquires the optimal AUC value for the Pavia image, and the GMMD method yields the largest AUC value for the Texas Coast image when the window size is 19. Therefore, the window size of guided filter is set as 5 for the Pavia image data and the simulated data, and the window size of guided filter is set as 19 for the Texas Coast data and the Los Angeles data.

For the sake of analyzing the effect of guided filtering rectification processing, we test the detection performance of the GMMD method with and without the guided filtering rectification processing on each experimental data set. Fig. 13(c) shows the AUC values of the GMMD method with and without the guided filtering processing, and the 2-D plots of the detection results on each data set are shown in Fig. 14. The GMMD method with and without the guided filtering processing is named GMMD_GF and GMMD_NGF, respectively. As shown in Fig. 13(c), for the simulated and Texas Coast images, the AUC values of GMMD_GF are slightly greater than those of GMMD_NGF, and GMMD_GF obtains the larger AUC values than GMMD_NGF for the Pavia and Los Angeles images. The subjective 2-D plots of each data set shown in Fig. 14 also have similar results, that is, the guided filtering step improves the detection performance of the proposed method and extrudes the anomaly pixels, especially for the Pavia and Los Angeles images.

We also compare the detection performance of each comparison method with and without the guided filtering rectification processing to further analyze the impact of guided filtering. The results of each method for the Texas Coast and Los Angeles images are shown in Fig. 15. The GRX, LRX, CRD, and LRaSMD methods with guided filtering are named GRX_GF, LRX_GF, CRD_GF, and LRaSMD_GF, respectively. From Fig. 15, it can be seen that guided filtering does not improve the performance of GRX and LRaSMD for both the Texas Coast and Los Angeles data. For Texas Coast data, guided filtering improves the detection performance of LRX and CRD, but the AUC values of LRX_GF and CRD_GF are lower than those of LRX and CRD for the Los Angeles data. These results show that the guided filtering processing is not necessarily beneficial to improve the detection performance of other comparison algorithms. As shown in Fig. 15, when the guided filter is applied to all methods, the GMMD method (GMMD_GF) also yields the largest AUC values. Fig. 14 shows that guided filtering improves the detection accuracy of the proposed GMM method, which reveals that guided filtering is indeed meaningful and effective for the GMM method. This is because the detection map constructed by the GMM-based weighting method may generate artifacts at the boundaries, and the other comparison methods do not need to consider this issue. The guided filtering step can avoid artifacts and make the neighboring pixels belonging to the same object tend to have similar values. The guided filtering step plays an important role in terms of improving

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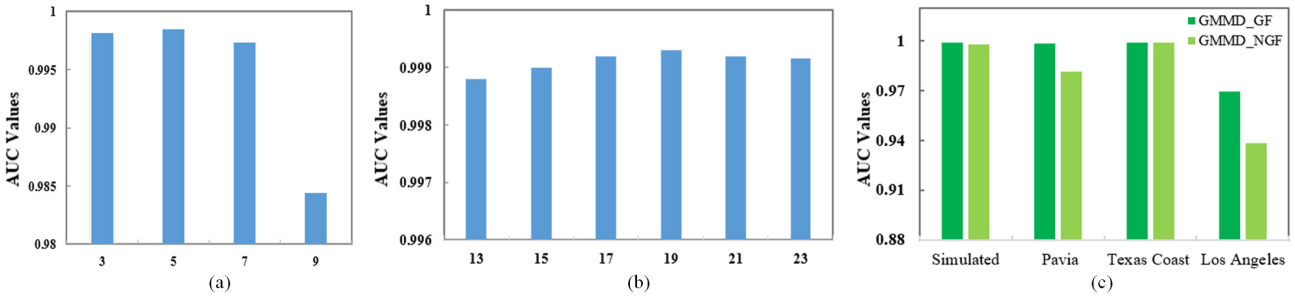


Fig. 13. (a) Detection results of the GMMD method with different window size settings for Pavia image scene. (b) Detection results of the GMMD method with different window size settings for Texas Coast image scene. (c) Performance of the proposed GMMD method with and without the guided filtering rectification processing.

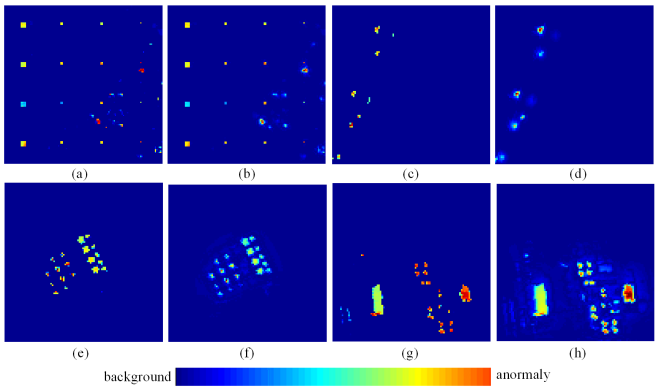


Fig. 14. 2-D plots of detection results of the proposed GMMD method with and without the guided filtering rectification processing. (a) GMMD_NGF (simulated image). (b) GMMD_GF (simulated image). (c) GMMD_NGF (Pavia image). (d) GMMD_GF (Pavia image). (e) GMMD_NGF (Texas Coast image). (f) GMMD_GF (Texas Coast image). (g) GMMD_NGF (Los Angeles image). (h) GMMD_NGF (Los Angeles image).

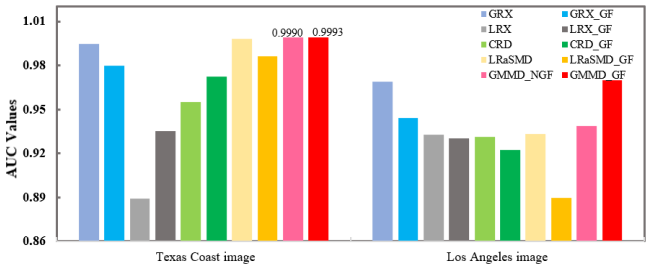


Fig. 15. Detection results of the guided filtering rectification processing for each competitive method.

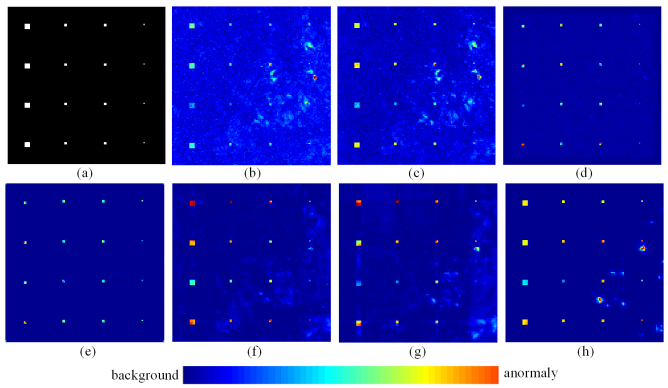


Fig. 16. 2-D plots of detection results of each method for the simulated data set. (a) Ground-truth map. (b) GRX. (c) GRX_DR. (d) LRX. (e) CRD. (f) LRaSMD. (g) LRaSMD_DR. (h) GMMD.

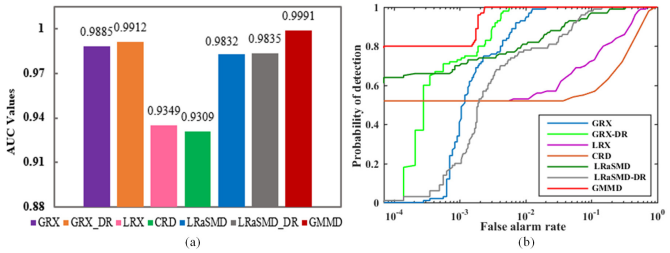


Fig. 17. AUC values and ROC curves of each method for the simulated data set.

the detection performance, and it is experimentally verified in Figs. 14 and 15.

F. Detection Performance of Each Method on Simulated Hyperspectral Data Set

The detection maps of each compared method on the simulated data set are shown in Fig. 16. The AUC values and ROC curves of each method are shown in Fig. 17. As shown in Fig. 16(b) and (c), GRX and GRX_DR detect most of the anomaly pixels, but they are incapable of suppressing the background pixels, which can also be reflected in the ROC curves shown in Fig. 17(b). When the false positive rate is small, the accuracy of GRX and GRX_DR is low. LRX and CRD successfully remove background information but extract insufficient anomalies. The ROC curves of LRX and CRD increase after a period of flatness in a detection probability. In contrast, LRaSMD, LRaSMD_DR, and GMMD that effectively highlight the anomaly targets achieve better detection performance compared with other methods, and GMMD removes more background information. It can also be seen from Fig. 17 that the proposed GMMD method achieves the highest AUC value, and the GMMD method exhibits higher true positive rates than other methods when the FAR varies from 0 to 1.

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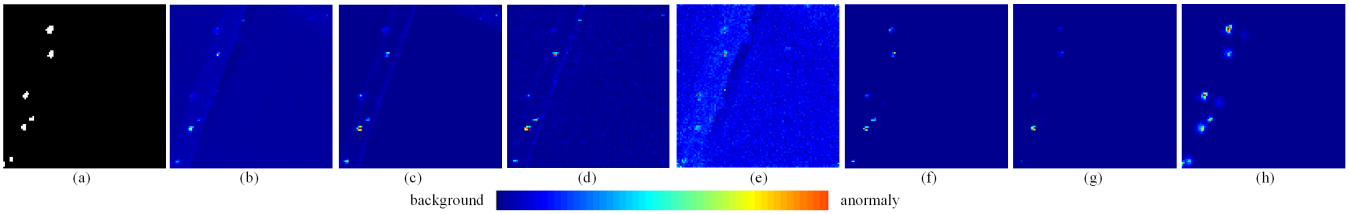


Fig. 18. 2-D plots of detection results of each method for Pavia image scene. (a) Ground-truth map. (b) GRX. (c) GRX_DR. (d) LRX. (e) CRD. (f) LRaSMD. (g) LRaSMD_DR. (h) GMMD.

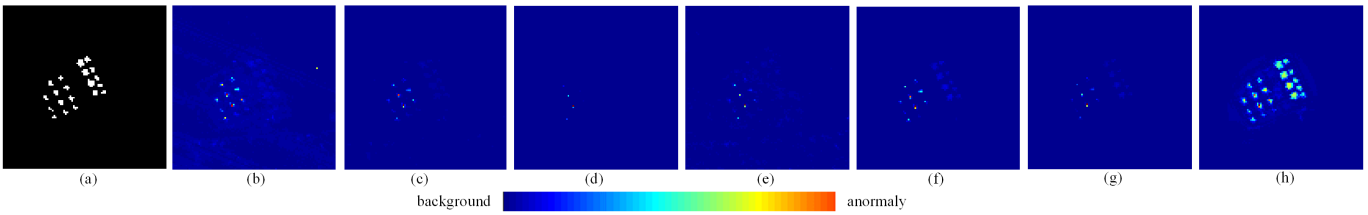


Fig. 19. 2-D plots of detection results of each method for Texas Coast image scene. (a) Ground-truth map. (b) GRX. (c) GRX_DR. (d) LRX. (e) CRD. (f) LRaSMD. (g) LRaSMD_DR. (h) GMMD.

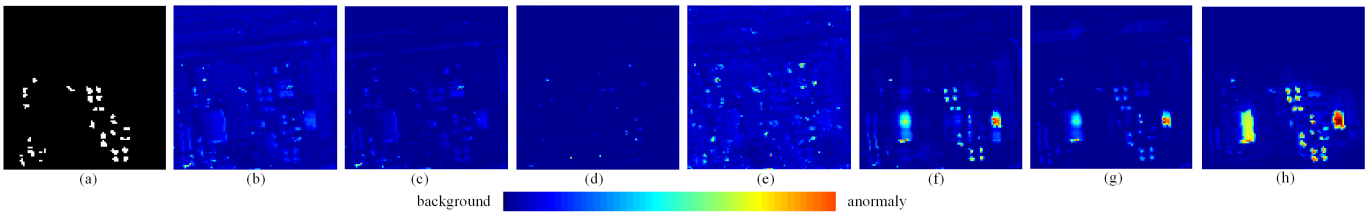


Fig. 20. 2-D plots of detection results of each method for Los Angeles image scene. (a) Ground-truth map. (b) GRX. (c) GRX_DR. (d) LRX. (e) CRD. (f) LRaSMD. (g) LRaSMD_DR. (h) GMMD.

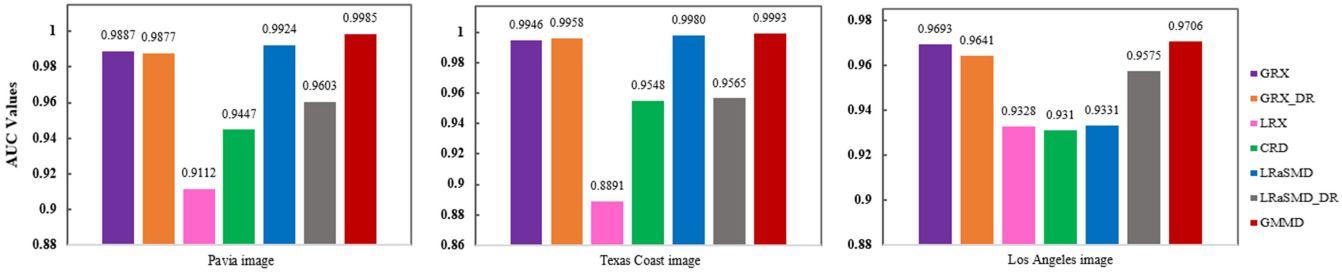


Fig. 21. AUC values of each method for three data sets.

G. Detection Performance of Each Method on Real Hyperspectral Data Sets

The detection maps of the compared methods and the proposed GMMD method for three real data sets are shown in Figs. 18–20, respectively. For the Pavia data set, as shown in Fig. 18, GRX, GRX_DR, and LRX detect some anomaly pixels but fail to suppress some background areas, such as the bridge. CRD shows a poor background suppression, which generates high FARs. LRaSMD and LRaSMD_DR obtain an acceptable background suppression but leave out certain abnormal pixels. By comparing with these methods, the proposed GMMD effectively extrudes the anomaly pixels and suppresses the background.

The visual detection results of each method for the Texas Coast HSI are shown in Fig. 19. Similar to the results of the Pavia data set, it is evident that GMMD performs the best. GRX and CRD fail to suppress the background. LRX detects few pixels and cannot effectively detect the abnormal targets, which results in poor detection performance. GRX_DR, LRaSMD, and LRaSMD_DR successfully suppress the background, but the anomaly pixels detected by them are not complete compared with the ground truth map. As shown in Fig. 20, the visual detection results of each method for the Los Angeles image scene are shown in a sequence. After analyzing these results, we can find that GRX and GRX_DR effectively detect the anomalies. However, they are not able to successfully remove the background and cause false

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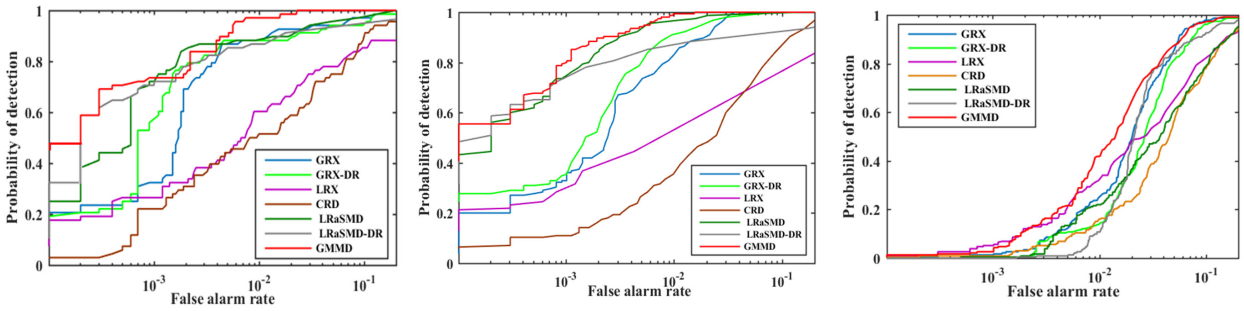


Fig. 22. ROC curves of each method for three data sets.

alarms. LRX only detects partial anomaly pixels, resulting in relatively low detection accuracy. CRD fails to highlight the anomalies from background, as shown in Fig. 20(e). LRaSMD, LRaSMD_DR, and GMMD extrude most of the anomalies. LRaSMD and LRaSMD_DR fail to suppress the background, and GMMD suppresses more background compared with LRaSMD and LRaSMD_DR.

To further accurately evaluate the detection performance of each method, the quantitative evaluation criteria are introduced. The AUC values and ROC curves of the compared methods and the proposed GMMD method for the real data sets are shown in Figs. 21 and 22, respectively. From left to right, the results in Figs. 21 and 22 are in turn the detection results of each method for the Pavia, Texas Coast, and Los Angeles data sets. From Fig. 21, it can be seen that GMMD yields the largest AUC values for three real data sets. As shown in Fig. 22, the proposed GMMD method obtains the best ROC curves for the Pavia and Texas Coast images, and the ROC curves generated by GMMD are the closest to the top-left corner among the experimental methods. The results obtained by LRX, CRD, and GRX are not satisfying since they fail to suppress the backgrounds or detect incomplete anomalies. For the Los Angeles data set, when the FAR is approximately 10^{-3}, the detection probability of LRX is slightly higher than GMMD, but the probability is low when the FAR is greater than 10^{-2}. In contrast, the ROC curve of GMMD shows better performance than other methods on the whole for the Los Angeles image. The above visual comparison and objective analysis validate that the proposed GMMD method is superior to other compared methods.

V. Conclusion

In this article, a novel GMM-based algorithm is proposed for anomaly detection in HSIs. The HSI is first partitioned into subsets, and the partitioned bands are fused to reduce the HSI dimension. Then, a GMM-based anomaly extraction method is proposed to effectively extrude the anomaly pixels of each band, the extracted results are fused by a GMM-based weighting method to construct an anomaly detection map, and this anomaly detection map is rectified by guided filtering to acquire the final detection map. Compared with other superior anomaly detection methods, the experiments on four hyperspectral data sets demonstrate that the proposed GMMD method provides better results and can effectively detect the anomalies with high accuracy.

References

[1] A. Chudnovsky, A. Kostinski, L. Herrmann, I. Koren, G. Nutesku, and E. Ben-Dor, "Hyperspectral spaceborne imaging of dust-laden flows: Anatomy of saharan dust storm from the Bodélé depression," *Remote Sens. Environ.*, vol. 115, no. 4, pp. 1013–1024, Apr. 2011.

[2] C. A. Bishop, J. G. Liu, and P. J. Mason, "Hyperspectral remote sensing for mineral exploration in Pulang, Yunnan province, China," *Int. J. Remote Sens.*, vol. 32, no. 9, pp. 2409–2426, May 2011.

[3] R. J. Ellis and P. W. Scott, "Evaluation of hyperspectral remote sensing as a means of environmental monitoring in the St. Austell China clay (kaolin) region, Cornwall, UK," *Remote Sens. Environ.*, vol. 93, nos. 1–2, pp. 118–130, Oct. 2004.

[4] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[5] Q. Wang, J. Lin, and Y. Yuan, "Salient band selection for hyperspectral image classification via manifold ranking," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 6, pp. 1279–1289, Jun. 2016.

[6] Q. Wang, Z. Meng, and X. Li, "Locality adaptive discriminant analysis for spectral–spatial classification of hyperspectral images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2077–2081, Nov. 2017.

[7] J. Qu, J. Lei, Y. Li, W. Dong, Z. Zeng, and D. Chen, "Structure tensor-based algorithm for hyperspectral and panchromatic images fusion," *Remote Sens.*, vol. 10, no. 3, p. 373, Mar. 2018.

[8] F. Zhang and K. Zhang, "Superpixel guided structure sparsity for multispectral and hyperspectral image fusion over couple dictionary," *Multimedia Tools Appl.*, vol. 79, no. 7, pp. 4949–4964, 2020.

[9] Y. Zhao, J. Yang, and J. C.-W. Chan, "Hyperspectral imagery super-resolution by spatial–spectral joint nonlocal similarity," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2671–2679, Jun. 2014.

[10] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[11] Y. Yuan, D. Ma, and Q. Wang, "Hyperspectral anomaly detection by graph pixel selection," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3123–3134, Dec. 2016.

[12] Z. Li, F. Nie, X. Chang, L. Nie, H. Zhang, and Y. Yang, "Rank-constrained spectral clustering with flexible embedding," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 6073–6082, Dec. 2018.

[13] H. Zhang and L. Cao, "A spectral clustering based ensemble pruning approach," *Neurocomputing*, vol. 139, pp. 289–297, Sep. 2014.

[14] P. Duan, X. Kang, S. Li, P. Ghamisi, and J. A. Benediktsson, "Fusion of multiple edge-preserving operations for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10336–10349, Dec. 2019.

[15] P. Duan, X. Kang, S. Li, and P. Ghamisi, "Noise-robust hyperspectral image classification via multi-scale total variation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 6, pp. 1948–1962, Jun. 2019.

[16] N. M. Nasrabadi, "Hyperspectral target detection: An overview of current and future challenges," *IEEE Signal Process. Mag.*, vol. 31, no. 1, pp. 34–44, Jan. 2014.

[17] I. S. Reed and X. Yu, "Adaptive multiple-band CFAR detection of an optical pattern with unknown spectral distribution," *IEEE Trans. Acoust., Speech, Signal Process.*, vol. 38, no. 10, pp. 1760–1770, Oct. 1990.

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[18] J. M. Molero, E. M. Garzón, I. García, and A. Plaza, "Analysis and optimizations of global and local versions of the RX algorithm for anomaly detection in hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 801–814, Apr. 2013.

[19] H. Kwon and N. M. Nasrabadi, "Kernel RX-algorithm: A nonlinear anomaly detector for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 388–397, Feb. 2005.

[20] S. Khazai, A. Safari, B. Mojaradi, and S. Homayouni, "An approach for subpixel anomaly detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 2, pp. 769–778, Apr. 2013.

[21] Q. Guo, B. Zhang, Q. Ran, L. Gao, J. Li, and A. Plaza, "Weighted-RXD and linear filter-based RXD: Improving background statistics estimation for anomaly detection in hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2351–2366, Jun. 2014.

[22] M. J. Carlotto, "A cluster-based approach for detecting man-made objects and changes in imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 2, pp. 374–387, Feb. 2005.

[23] S. Matteoli, T. Veracini, M. Diani, and G. Corsini, "A locally adaptive background density estimator: An evolution for RX-based anomaly detectors," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 323–327, Jan. 2014.

[24] W. Li and Q. Du, "Collaborative representation for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 3, pp. 1463–1474, Mar. 2015.

[25] J. Li, H. Zhang, L. Zhang, and L. Ma, "Hyperspectral anomaly detection by the use of background joint sparse representation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2523–2533, Jun. 2015.

[26] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Simultaneous joint sparsity model for target detection in hyperspectral imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 4, pp. 676–680, Jul. 2011.

[27] X. Dong *et al.*, "Weighted locality collaborative representation based on sparse subspace," *J. Vis. Commun. Image Represent.*, vol. 58, pp. 187–194, Jan. 2019.

[28] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis," *J. ACM*, vol. 58, no. 3, p. 11, 2011.

[29] Y. Xu, Z. Wu, J. Li, A. Plaza, and Z. Wei, "Anomaly detection in hyperspectral images based on low-rank and sparse representation," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 4, pp. 1990–2000, Apr. 2016.

[30] W. Sun, C. Liu, J. Li, Y. M. Lai, and W. Li, "Low-rank and sparse matrix decomposition based anomaly detection for hyperspectral imagery," *J. Appl. Remote Sens.*, vol. 8, no. 1, pp. 152–160, May 2014.

[31] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York, NY, USA: Springer, 2006.

[32] A. Berge and A. H. S. Solberg, "Structured Gaussian components for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3386–3396, Nov. 2006.

[33] M. A. T. Figueiredo and A. K. Jain, "Unsupervised learning of finite mixture models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 3, pp. 381–396, Mar. 2002.

[34] B. A. S. Hasan and J. Q. Gan, "Sequential EM for unsupervised adaptive Gaussian mixture model based classifier," in *Machine Learning and Data Mining in Pattern Recognition*. Berlin, Germany: Springer, 2009.

[35] X. He, D. Cai, Y. Shao, H. Bao, and J. Han, "Laplacian regularized Gaussian mixture model for data clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 23, no. 9, pp. 1406–1418, Sep. 2011.

[36] W. Li, S. Prasad, and J. E. Fowler, "Hyperspectral image classification using Gaussian mixture models and Markov random fields," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 1, pp. 153–157, Jan. 2014.

[37] M. A. Kerroum, A. Hammouch, and D. Aboutajdine, "Textural feature selection by joint mutual information based on Gaussian mixture model for multispectral image classification," *Pattern Recognit. Lett.*, vol. 31, no. 10, pp. 1168–1174, Jul. 2010.

[38] H. Greenspan, A. Ruf, and J. Goldberger, "Constrained Gaussian mixture model framework for automatic segmentation of MR brain images," *IEEE Trans. Med. Imag.*, vol. 25, no. 9, pp. 1233–1245, Sep. 2006.

[39] T. M. Nguyen and Q. M. J. Wu, "Fast and robust spatially constrained Gaussian mixture model for image segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 23, no. 4, pp. 621–635, Apr. 2013.

[40] Y. Dong, T. Long, W. Jiao, G. He, and Z. Zhang, "A novel image registration method based on phase correlation using low-rank matrix factorization with mixture of Gaussian," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 446–460, Jan. 2018.

[41] K. He, J. Sun, and X. Tang, "Guided image filtering," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 6, pp. 1397–1409, Jun. 2013.

[42] J. Qu, Y. Li, and W. Dong, "Hyperspectral pansharpening with guided filter," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 2152–2156, Nov. 2017.

[43] N. Billor, A. S. Hadi, and P. F. Velleman, "BACON: Blocked adaptive computationally efficient outlier nominators," *Comput. Statist. Data Anal.*, vol. 34, no. 3, pp. 279–298, Sep. 2000.

[44] X. Kang, X. Zhang, S. Li, K. Li, J. Li, and J. A. Benediktsson, "Hyperspectral anomaly detection with attribute and edge-preserving filters," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 10, pp. 5600–5611, Oct. 2017.

[45] C. Ferri, J. Hernández-Orallo, and P. Flach, "A coherent interpretation of AUC as a measure of aggregated classification performance," in *Proc. Int. Conf. Mach. Learn.*, 2011, pp. 657–664.

[46] J. Kerekes, "Receiver operating characteristic curve confidence intervals and regions," *IEEE Geosci. Remote Sens. Lett.*, vol. 5, no. 2, pp. 251–255, Apr. 2008.

[47] G. Schwartz, "Estimating the dimension of a model," *Ann. Stat.*, vol. 6, no. 2, pp. 31–38, 1978.

[48] C. Fraley and A. E. Raftery, "Model-based clustering, discriminant analysis, and density estimation," *J. Amer. Stat. Assoc.*, vol. 97, no. 458, pp. 611–612, 2002.



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